

In this Standard the comma has been adopted as decimal sign.

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1. SCOPE

This Standard establishes the leak-detection system procedures used for detecting leaks in aircraft systems, or aircraft structures, except oxygen, hydraulic and pneumatic systems.

2. APPLICATION

The Helium leak detecting is applied on fuel system and tanks inspection, for all types of aircrafts.

3. APPLICABLE DOCUMENTS

Embraer Standards

MEP 18-028 - Aluminum Adhesive Tape.

NE 40-134 - Sealing of Aircraft - General Requirements.

Embraer Procedures

doc.emb 247 - "Treinamento - Qualificação e Certificação de Pessoal".

NOTE

The doc.emb procedure is for Embraer use only.

Foreign Standards

Military Specification (DoD - USA)

MIL-PRF-25567 - Leak Detection Compound, Oxygen Systems.

MIL-PRF-27407 - Propellant Pressurizing Agent, Helium.

Occupational Safety and Health Administration (OSHA)

OSHA 1910.1200 - Hazard Communication Standard.

4. DEFINITIONS

4.1 LEAK-PATH MEDIUM

The leak-path medium is a material that will follow a leak-path and cause a detectable change in the interior.

4.2 INDICATOR

The indicator is a material or instrument that will indicate the presence of the leak-path medium.

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4.3 LEAK DETECTION SOLUTION

Solution with very low surface tension that will form bubbles when spread over leaks. The leak detection solution will locate the leak by the evidence of bubbles shall not react with the testing objects materials.

4.4 GROSSE LEAKS

Leaks that have a leak rate greater than 1×10^{-3} mbar.l/s.

4.5 SMALL LEAKS

Leaks that have a leak rate smaller than 1×10^{-3} mbar.l/s and greater than 1×10^{-5} mbar.l/s.

4.6 TESTING OBJECT

Tank or system which need to have its tightness proven according to the specified parameters. Every external contact of the testing object shall be available for blanking.

4.7 PROBE

Equipment that sniffs the tracer fluid, collecting and conducting it to the detection unit, which gives out a signal.

4.8 TESTING SPEED

It is the speed the operator shall use to check the testing object. If operator goes to fast, no leak will be detected; if operator goes to slow, a leak greater than the real one will be detected. To get the best testing speed, the operator shall use a calibration leak.

4.9 HELIUM FLUID

Is a colorless, odorless, tasteless, non-toxic, inert monatomic chemical element fluid. In this inspection, the helium fluid may be used mixed with air or pure according to below.

4.9.1 Mixed (50% Helium and 50% Air)

The mixture (helium and air) inside the testing object shall be homogeneous.

The minimum concentration is helium (50% He) mixed with air (50% air).

The testing personnel shall ensure that every extremity of the testing object is filled with the fluid.

4.9.2 Pure (Helium)

The fluid inside the testing object shall be only helium (not mixed with air).

The testing personnel shall ensure that every extremity of the testing object is filled with the fluid.

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4.10 CALIBRATION LEAK

It is an instrument which allows a constant leak rate of a determined fluid.

The leak rate is specified by the supplier of the calibration leak and depends on temperature. This instrument may be used to calibrate leak testing equipment and determining the optimal testing speed.

4.11 MAXIMUM ADMISSIBLE LEAK RATE

It is the maximum leak rate detectable by the leak detecting systems that will not endanger the operation of the system or the aircraft. For the systems contemplated by this procedure, the maximum admissible leak rate is 1×10^{-5} mbar.l/s.

4.12 CRITICAL POINTS

Points or areas on the testing object that are more susceptible to leaks than other or that may cause more damage when a leak occurs. The Critical points shall be determined for each of the testing objects, preferably on sketch.

5. PERSONNEL QUALIFICATION

The personnel executing the leak test procedures shall be qualified according to the procedure of Embraer Personnel Qualification (doc.emb 247 procedure).

6. ENVIRONMENT

To make sure that the testing of the object is done on its best conditions, the testing personnel shall ensure that:

6.1 There is ventilation on the area, so that once the tracer fluid is evacuated from the testing object, there will be no fluid accumulation.

6.2 The final helium concentration in the tanks was 5%, which accommodates the OSHA minimum of 19,5% oxygen in a confined space.

7. TEST DETECTION METHODS

7.1 PRESSURE BACK DROP (GROSSE LEAK)

7.1.1 After configuring and testing the free flow, the system shall be pressurized until stabilization at the set value, allowing a tolerance of up to 5% of the nominal pressure. Once stabilized, monitor for 10 to 15 minutes as per procedure. During this period, the pressure drop shall not exceed 10% of the nominal stabilization value. Excessive pressure loss (greater than 10%) indicates a gross leak, which may prevent the use of other detection methods and lead to complete depressurization and fluid loss.

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Table 1 shows aircraft types, regions and pressures used in the leakage test.

TABLE 1 : AIRCRAFT TYPES X REGIONS X PRESSURES

AIRCRAFT TYPE		REGION	PRESSURE (±5% psi)
EMB-145	LEGACY (600/650)	TANK, AFT	2,0
		TANK FUSELAGE LINES	50,0
		TANK DIF LINES	50,0
		FUEL, FUSELAGE/STUB LINES	50,0
		TANK AFT, PRESSURE SYSTEM	2,5
		TANK AFT, ACCESS BOX INST	1,0
		WING STUB STRUCTURE	2,5
		VENTRAL TANK STRUCTURE	2,5
EMB-550	LEGACY (450/500)	WING STUB STRUCTURE/ WING STRUCTURE	3,0
		MOTIVE FLOW LINE	20,0
		SHROUD LINE	2,5
		FEED LINE	20,0
		TRANSFER LINE	20,0
		VENT LINE	2,5
		FIREX	20,0
		AUXILIARY FUEL TANKS (VENTRAL AND STUB ANNEX)	9,5
ERJ 170/190 (EJet's E1 170/175/190/195)		FIREX	50
		WING/STUB STRUCTURE (1)	3,0
		FUEL, AFTER CLOSING	2,5
		FUEL LINE (SUPPLY/FEED)	20,0
		MOTIVE FLOW LINE	50,0
		VENTILATION LINE	5,0
ERJ 190 (EJet's E2 175/190/195)		FIREX	50,0
		WING/STUB STRUCTURE (1)	3,0
		FUEL, AFTER CLOSING	2,0
		FUEL LINE (SUPPLY/FEED/OBGIS/ TRANSFER/VENTILATION)	25,0
		MOTIVE FLOW LINE	25,0
ERJ 190 (LINEAGE)		FWD FUEL TANK EQUIP MOD A1	2,0
		FWD FUEL TANK EQUIP MOD A2	2,0
		FWD FUEL TANK EQUIP MOD E1	2,0
		FWD FUEL TANK EQUIP MOD E2	2,0
		FWD FUEL TANK EQUIP MOD E3	2,0
		FWD FUEL TANK EQUIP MOD E4	2,0
		AFT FUEL TANK EQUIP MOD D1	2,0
		AFT FUEL TANK EQUIP MOD D2	2,0
		AFT FUEL TANK EQUIP MOD D3	2,0
		AFT FUEL TANK EQUIP MOD D4	2,0
		AFT FUEL TANK EQUIP MOD C	2,0
		AFT FUEL TANK EQUIP MOD F	2,0
		VERIFICATION BETWEEN BARRIERS	1,0
		AFT FUEL TANK EQUIP MOD G	2,0
		WING/STUB STRUCTURE	3,0
EMB-500/505 (PHENOM 100/300)		FIREX	20,0
		SHROUD LINE	2,5
		VENT LINE	2,5
		FUEL LINES	20,0
		FUEL TANK LEAKAGE	2,5
EMB-314 ALX (SUPER TUCANO)		UNDERWING TANK LEAKAGE	2,0
		FEED FUEL ENGINE SYSTEM INSTALLATION	50,0
		FUEL SYSTEM LINE LEAKAGE TEST	50,0
		WING TANK FUEL TEST	2,0
		TEST, LEAKAGE, WING FUEL TANK LINE	50,0
		FEED FUEL FUSELAGE TANK INSTALLATION	50,0
		LEFT HALF WING STRUCTURE – COMPL.	2,5
		RIGHT HALF WING STRUCTURE – COMPL.	2,5

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TABLE 1 : AIRCRAFT TYPES X REGIONS X PRESSURES

AIRCRAFT TYPE	REGION	PRESSURE (±5% psi)
EMB-390 (KC-390)	HALF WING / WING CENTRAL TANK STRUCTURE	2,5
	AFFT, FUEL TANK	1,5
	AFFT, VERIFICATION BETWEEN BARRIERS	2,0
	FEED, REFUEL AND RETURN MAIN LINES	25,0
	TRANSFER LINE LH	25,0
	TRANSFER LINE RH	25,0
	VENT LINE	2,5
	FIREX	25,0
GRIPEN	WING STUB STRUCTURE / WING STRUCUTURE	7,25 ±0,15
	FEED LINE	6,5 ±0,15
	TRANSFER LINE	7,25 ±0,15 (WING) 6,5 ±0,15 (FUSELAGE)
	VENT LINE	6,5 ±0,15
	FIREX	NA
IPANEMA	WING STUB STRUCTURE / WING STRUCUTURE	NA
	FEED LINE	NA
	TRANSFER LINE	NA
	VENT LINE	NA
	FIREX	NA
	TANK	1,5

(1) The leakage test shall be performed using the pressure defined by Table 1 and shall be comply with the following requirements:

- 1) The missing fasteners shall not exceed 10% of the fasteners joint (bay).
- 2) Shall be examined for evidence of a continuous bead of faying surface sealant squeezed out to ensure an adequate sealing, without gaps.
- 3) Not allowed more than two fasteners missing in sequence.

Attending the above requirements the leakage test may be performed. The adhesive tape (MEP 18-028) shall be applied in all holes. Otherwise, Product Engineering disposition is required.

NOTE

All pending issues that prevent the execution of the test shall be evidenced in a document of non-conformity (rework).

The leakage test located on fasteners installed after leakage test of structure may be performed with the pressure defined to "FUEL, AFTER CLOSING".

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7.1.2 For the pressure back drop method, there are some characteristics that shall be observed, in order to make the test accordingly.

The testing time shall be carefully calculated, using the volume of the testing object and the maximum admissible leak rate. It depends on the admissible leak rate and on the sensitivity of the pressure measuring systems.

$$\Delta t = \Delta(p) \cdot V / Q_L$$

Where:

Q_L = the admissible leak rate (mbar.l/s) according to item 4.11.

$\Delta(p)$ = the sensitivity of the pressure measuring system (mbar).

V = the volume of the testing object (l). The volume of each system shall be individually calculated by Product Development Engineering. Product Manufacturing Engineering shall define one or more test object volume setup by combining the individually calculated system volumes.

Δt = the testing time (s) ⁽¹⁾ ⁽²⁾

(1) The test time shall be performed between 30 to 60 minutes after injection of fluid into compartment. If the time exceeds 60 minutes, empty the compartment and begin the test again.

(2) The calculation shall be executed by Product Manufacturing Engineering and properly reported in the Process Instructions.

7.2 LEAK DETECTION SOLUTION

For the soap solution leak-testing method, there are some characteristics that shall be observed, in order to make the test accordingly.

7.2.1 The critical points shall be covered with soap solution that does not react with the components of the system.

7.2.2 The personnel shall ensure that all the points may be easily accessed with eye contact.

7.2.3 There shall be light available.

7.2.4 The evidence of leak will be the presence of bubbles.

7.2.5 If the leak is too large, the soap solution will be diverted from the actual leak spot, making it impossible to locate it.

7.3 HELIUM LEAK TEST

This method shall be used to locate very small leaks. The tests consist of pressurizing the testing object (see Table 1) with a fluid (pure or mixed with air), according to item 4.9 and searching it with help of the probe. An alarm will sound once a leak is found.

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7.3.1 General Test Procedures

To make the testing safer and to lose a smaller amount of leak path medium, the testing shall be done according to the following steps:

- a) pressurize (see Table 1) the object with the operating pressure;
- b) search for grosse leaks with leak detection solution;
- c) if leaks are found, they shall be repaired and the exact location shall be reported (preferably on sketch);
- d) this test with leak detection solution shall be repeated until no more leaks are found;
- e) depressurize the object carefully;
- f) pressurize (see Table 1) the tank with the fluid according to item 4.9;
- g) check that extreme points of the object have been pressurized with the mixture by purging (the purging points for each object shall be indicated on sketch or drawing);
- h) search for small leaks with the leak indicator (Heli-Test);
- i) if leaks are detected, eliminate the leak and continue the search until no more leaks are found;
- j) repair the leaks and report the exact location (preferably on sketch); and
- k) reapply the test to check the repaired areas.

7.3.2 Special Test Procedure

7.3.2.1 Shrouded Tubes/Pipes

For the second barrier of shrouded tubes/pipes (external barrier) only, the pressure test shall be carried out according to the following steps:

- a) calculate the test time according to item 7.1.2;
- b) if the calculated time is feasible for performing the pressure drop test, pressurize the test object with air according to Table 1, otherwise, follow all steps of the General Test Procedure (item 7.3.1);
- c) wait test object pressure stabilization;
- d) verify whether a pressure drop occurs during the previously calculated test time in accordance with the drop pressure tolerance;

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e) if no pressure drop is detected, carefully depressurize the test object and end the test procedure, otherwise, search for leaks using a leak detection solution according to item 7.2;

f) if leakage is detected, it shall be repaired and properly reported;

g) repeat the pressure drop test with leak detection solution until no more leaks are found; and

h) if the pressure drop persists and leak detection with soap solution is no longer feasible, apply item “f” and subsequent steps of the General Test Procedure in full (item 7.3.1).

NOTE

Primary barrier of shrouded tubes/pipes (internal tube) shall be tested according to item 7.3.1.

8. MATERIALS

8.1 INDICATOR

There are two possible indicators:

8.1.1 Leak Detection Solution

To be satisfactory, the leak-testing soap solution shall be attending the MIL-PRF-25567 standard.

8.1.2 Helium

To be satisfactory, the test fluid shall be non-toxic, not ozone depleting, non-flammable, not detrimental to common materials, and inexpensive.

Pure helium or mixed (50% He with 50% air), minimum, shall be used for the test. The helium fluid shall comply with MIL-PRF-27407 requirements.

8.2 LEAK-PATH MEDIUM

This nondestructive test method covers procedures for testing and locating the sources fluid leaking at the rate of $1,0 \times 10^{-5}$ mbar.l/s.

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9. REQUIREMENTS

9.1 TECHNICAL REQUIREMENTS

9.1.1 Sensitivity

The instrument shall give easily detectable evidence of leakage within 15 seconds.

Determinations shall be made in a draft free room maintained at ambient temperature.

Normal laboratory lighting conditions shall be used unless special lighting is specified by the supplier. When applicable, special lightning requirements will be used after Materials Engineering approval.

9.1.2 Calibration

The leak detectors used in making leak tests by these test methods are not calibrated in the sense that they are taken to the standards laboratory, calibrated, and then returned to the job. Rather, the leak detector is used as a comparator between a leak standard (set to the specified leak size) which is part of the instrumentation, and the unknown leak. However, the sensitivity of the leak detector is checked and adjusted on the job so that a leak of specified size will give a readily observable, but not off-scale reading.

To verify sensitivity, reference to the leak standard shall be made before and after a prolonged test. When rapid repetitive testing of many items is required, refer to the leak standard often enough to ensure that desired test sensitivity is maintained.

9.1.3 Interferences

The atmosphere contains about five parts per million (ppm) of helium, which is being continuously drawn in by the detector probe. This background shall be "zeroed out" before leak testing using helium may proceed. Successful leak testing is contingent on the ability of the detector to discriminate between normal atmospheric helium, which is very constant, and an increase in helium due to a leak. If the normally stable atmospheric helium level is increased by release of helium in the test area, the reference level becomes unstable, and leak testing more difficult.

Dirt and Liquids - As the orifice in the detector probe is very small, the parts being tested shall be clean and dry to avoid plugging. Check the cleaning procedure on the equipment manual.

Helium out gassed from Absorbent Materials - Helium absorbed in various nonmetallic materials (such as rubber or plastics) may be released during the test. If the rate and magnitude of the amount released approaches the amount released from the leak, the reliability of the test is decreased.

NOTE

For rubber tanks, a research shall be done to determine the helium permeation rate for that case, before the procedures are written.

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9.2 PROCESS REQUIREMENTS

Procedures and monitoring processes are being implementing to ensure that inert, odorless helium does not dangerous or asphyxiating material to operators in confined areas.

The helium (pure or mixed with air), according to specified in item 4.9, from the tank may be exhausted directly into the work area after the test. The brief helium ventilation also enables mechanics to enter the tank to complete necessary repairs.

NOTES

1) Care shall be taken to provide sufficient ventilation to prevent increasing the helium background and the percentage of the leaking trace fluid captured by the probe.

2) Results are limited by the helium background and the percentage of the leaking trace fluid captured by the probe.

10. GENERAL NOTES

10.1 DRAWING CALLOUT

This Standard shall be called out on drawings through the following note:

NI 981 - Perform helium leak detecting test, according to NE 07-024.

-----/ANNEX 1

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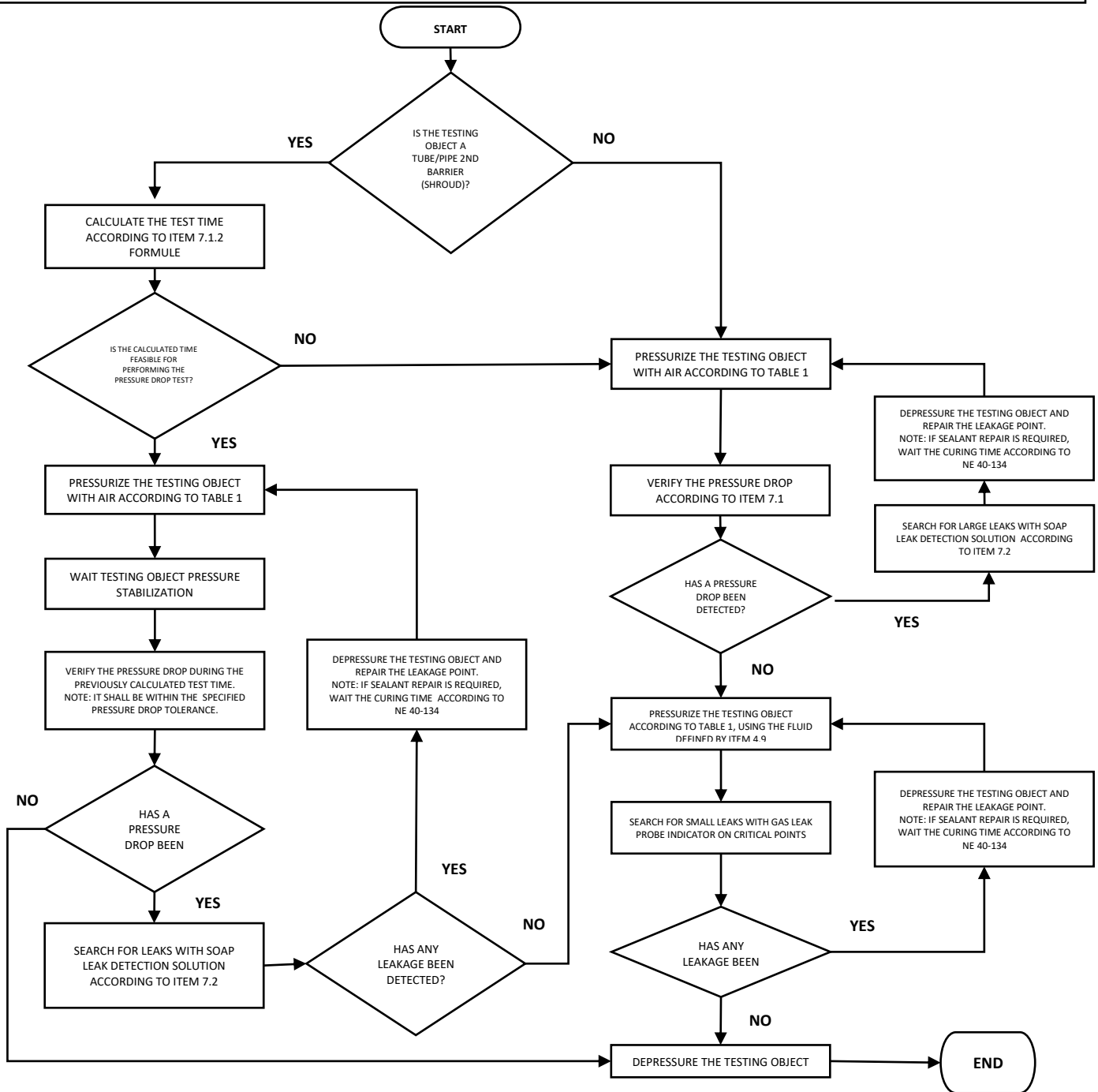
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ANNEX 1 : SCHEMATIC PROCESS LEAK DETECT PROCESS



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